LESSON PLAN

EHM (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of Wire locking, Practical on wire locking procedure
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of Wire locking, Practical on wire locking procedure
5. Specific Objectives	28.1.3.1 Trainee will be able to explain Types of Wire locking, wire locking procedure.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration and practice of wire locking procedure using standard hardware; Safety precautions include wearing PPE and ensuring correct tension; Instructor supervision required for all trainee attempts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of wire locking in aircraft structures?
(b) Name two common types of wire locking methods used in aviation.
(c) Why is wire locking preferred over other locking methods for certain fasteners?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and purpose of wire locking • Types of wire locking: Single and Double • Selection criteria for wire locking application • Material specifications for locking wires • Standard wire locking procedure steps	L & Ds	PPT, IB	15 min
(b)	• Step-by-step wire locking demonstration • Correct tensioning and twisting techniques • Inspection criteria for completed locks • Common errors and their prevention • Safety precautions during wire locking	L & Ds	PPT, PDF	20 min
(c)	• Trainee practice on wire locking procedure • Instructor feedback and correction • Evaluation of wire locking quality • Documentation of wire locking records • Review of key learning points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between single and double wire locking methods.
(b) Describe the critical steps in ensuring proper wire tension during locking.
(c) List three safety precautions that must be observed during wire locking.

12. Home Assignment and Follow Up	Prepare a brief report on the wire locking procedure covered in this lesson and identify one potential failure mode if the procedure is not followed correctly, referencing LP 02 for context on fastener integrity.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Torque tightening procedure - calculation of torque, torque tightening practice and calibration procedure of torque wrench. Practical on torque tightening procedure	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min